

Introduction to Pinch Analysis and Heat Exchanger Network Design

From Process Stream Data to the First Grid Diagram

Ada Siepmann

B.A. in Chemistry, Mount Holyoke College

B.S. Candidate in Chemical Engineering, UMass Amherst

This document originated from personal lecture notes on Pinch Analysis and Heat Exchanger Network Design, prepared during a dual-degree course of study in Chemistry and Chemical Engineering. The material was subsequently reorganized, expanded, and formalized into the present introduction.

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Introduction and Motivation

Heat is one of the most valuable resources in industrial processes. In many production plants, some process streams must be cooled while others must be heated. Viewed separately, these requirements are often satisfied by external cooling and heating utilities such as cooling water, refrigeration systems, steam, or fired heaters. However, from an energy perspective, this approach is inefficient because heat that is removed from one part of the process may simultaneously be required somewhere else within the same plant.

The motivation for studying heat integration is both economic and environmental. Every unit of heat that can be recovered internally reduces the demand for external heating and cooling utilities. Consequently, improved heat integration may lower operating costs, reduce fuel consumption, decrease cooling requirements, and reduce the associated environmental impact. The challenge is therefore not merely to identify isolated opportunities for heat recovery, but to determine how much heat can be recovered throughout the process as a whole.

The central idea of heat integration is simple: before purchasing additional heat or rejecting valuable heat to the environment, one should first investigate how much heat can be exchanged internally between the process streams themselves. In practice, this question quickly becomes difficult. Even moderately sized processes may contain dozens of hot and cold streams, and the possible combinations between these streams soon become too numerous to be analysed by intuition alone.

Pinch Analysis was developed to address precisely this problem. Rather than starting with the design of individual heat exchangers, the method first determines the thermodynamic limits of heat recovery. It answers questions such as:

- How much external heating is truly required?
- How much external cooling is unavoidable?
- Where is the thermodynamic bottleneck of the process?
- How should the process be divided before a heat exchanger network is designed?

The result of this first stage is the so-called *Pinch Point*, which separates the process into a heat-deficit region and a heat-surplus region. Once the pinch has been identified, the engineer can proceed to the actual design of the Heat Exchanger Network (HEN).

It is important to distinguish between Pinch Analysis and Heat Exchanger Network Design. Pinch Analysis determines the thermodynamic targets of the process, whereas Heat Exchanger Network Design determines how these targets can be approached using actual heat exchangers and utilities.

The material presented in this document originated from personal notes taken during a lecture on Pinch Analysis and Heat Exchanger Network Design. After the lecture, these notes were reviewed, reorganized, and formalized into a coherent summary of the methodology. During this process, a small supporting workbook, `HEN_Pinch_Analysis_Helper.xlsm`, was developed to reproduce the individual steps of a pinch study, validate example calculations, and provide a practical framework for exploring the concepts discussed in the notes.

The overall methodology can be viewed as four consecutive steps:

1. Collect the process stream data.
2. Perform the pinch analysis and identify the pinch point.

3. Construct a preliminary heat exchanger network using the pinch design rules.
4. Refine, validate, and further develop the resulting network.

The first two steps are largely thermodynamic in nature and lead to the energy targets of the process. The third step introduces engineering judgement and network synthesis principles such as stream matching, heat-capacity flow-rate considerations, and stream splitting. For large industrial systems, these activities are typically supported by specialized process integration software, but the underlying concepts can be understood using comparatively small examples.

To support the learning process, the accompanying workbook `HEN_Pinch_Analysis_Helper.xlsx` was created. The workbook assists in collecting stream data, generating shifted temperatures, constructing the heat cascade, identifying the pinch point, and creating a first grid diagram. It is intended as an educational and validation tool rather than a fully automated heat exchanger network synthesis package.

The remainder of this document follows the logical development of a pinch study. First, the required process data are introduced. Next, the pinch analysis itself is explained using simple examples and intuitive reasoning. Finally, the connection between the pinch results and the construction of a heat exchanger network is discussed.

1 Collecting the Process Data

1.1 Hot and Cold Process Streams

Every pinch study begins with the identification of the relevant process streams. From the perspective of heat integration, two types of streams are distinguished.

A *hot stream* is a process stream that must be cooled during the process. Such a stream releases heat and therefore represents a potential heat source for other parts of the plant.

A *cold stream* is a process stream that must be heated during the process. Such a stream absorbs heat and therefore represents a potential heat sink.

The terms *hot* and *cold* should be understood relative to the desired process change rather than the absolute temperature level of the stream. A stream is classified as hot if it must be cooled, whereas a stream is classified as cold if it must be heated.

For example,

- a stream cooling from 220 °C to 80 °C is a hot stream,
- whereas a stream heating from 20 °C to 180 °C is a cold stream.

The objective of heat integration is to transfer as much heat as possible from the hot streams to the cold streams before external heating or cooling utilities are employed.

1.2 Required Stream Data

Once the hot and cold streams have been identified, the relevant process data must be collected. For each stream, four quantities are required:

$$T_s, \quad T_t, \quad \dot{m}, \quad c_p. \quad (1)$$

Here

- T_s denotes the supply temperature,
- T_t denotes the target temperature,
- $\dot{m}$ denotes the mass flow rate,
- c_p denotes the specific heat capacity.

Typical engineering units are

$$T_s, T_t : ^\circ\text{C}, \quad (2)$$

$$\dot{m} : \text{kg/s}, \quad (3)$$

$$c_p : \text{kJ}/(\text{kg} \cdot \text{K}). \quad (4)$$

The temperatures define the required thermal change of the stream, whereas the mass flow rate and heat capacity determine how much thermal energy is associated with this change.

The quality of the pinch analysis can never exceed the quality of the underlying stream data. Consequently, careful data collection is one of the most important steps of any pinch study.

1.3 Heat-Capacity Flow Rate

Although the mass flow rate and the specific heat capacity are usually collected separately, pinch analysis is largely performed in terms of their product, the *heat-capacity flow rate*:

$$CP = \dot{m} c_p. \quad (5)$$

The quantity CP measures the rate at which thermal energy changes with temperature. A stream with a large CP undergoes a relatively small temperature change for a given amount of transferred heat, whereas a stream with a small CP experiences a larger temperature change under the same heat load.

For example, consider a stream with

$$\dot{m} = 1.0 \text{ kg/s} \quad (6)$$

and

$$c_p = 2.0 \text{ kJ}/(\text{kg} \cdot \text{K}). \quad (7)$$

Its heat-capacity flow rate is

$$CP = 2.0 \text{ kW/K}. \quad (8)$$

In practice, engineers often discuss streams primarily in terms of their CP values because these directly determine how much thermal energy is associated with a given temperature interval.

1.4 Example Stream Table

To illustrate the required input data, consider the simple example shown in Table 1. The same example will be used throughout the subsequent sections when introducing the pinch analysis itself.

Table 1: Example process stream data.

Stream	T_s (°C)	T_t (°C)	$\dot{m}$ (kg/s)	c_p (kJ/(kg · K))	CP (kW/K)
H1	220	80	1.0	2.0	2.0
H2	180	60	1.0	1.5	1.5
C1	20	180	1.0	1.5	1.5
C2	50	150	1.0	2.0	2.0

At this stage the streams are merely described individually. No attempt has yet been made to connect hot and cold streams or to recover heat between them. The table simply contains the information required to begin the analysis.

The next step is to determine how much heat can theoretically be recovered within the process. This requires the introduction of a minimum temperature difference, shifted temperatures, temperature intervals, and ultimately the heat cascade.

2 Pinch Analysis

The process data collected in Section 1 describe the individual heating and cooling requirements of the process streams. They do not yet reveal how much heat can be exchanged internally or how much external heating and cooling will ultimately be required.

This is the purpose of Pinch Analysis. Before individual heat exchangers are selected, the method determines the thermodynamic limits of heat recovery. It identifies the minimum external heating and cooling requirements and locates the pinch point, which represents the principal thermodynamic bottleneck of the process.

The analysis proceeds through a sequence of conceptually simple steps. First, a minimum allowable temperature difference is specified. The stream temperatures are then shifted so that this restriction is automatically incorporated into the subsequent calculations. The shifted temperatures are sorted into temperature intervals, an energy balance is established for each interval, and the resulting balances are combined in the heat cascade.

2.1 Why a Minimum Temperature Difference is Required

Heat can flow spontaneously only from a higher temperature to a lower temperature. A temperature difference is therefore required to drive heat transfer inside a heat exchanger.

Consider two process streams that both have a temperature of 100 °C. Even if one stream is intended to heat the other, no heat transfer can occur because there is no temperature difference between them. A finite temperature difference is therefore essential for practical heat exchange.

In principle, the temperatures of a hot and a cold stream could be allowed to approach each other very closely. This would increase the amount of heat that can be recovered internally. However, a small temperature difference also reduces the driving force for heat transfer. A larger

heat-transfer area is then required to achieve the same heat duty. In the limiting case of a vanishing temperature difference, the required heat-transfer area would become unbounded.

Pinch Analysis accounts for this practical limitation by introducing a minimum allowable temperature difference,

$$\Delta T_{\min}. \quad (9)$$

The value of $\Delta T_{\min}$ is a design parameter rather than a universal physical constant. Its selection depends on the fluids, the process conditions, the available heat-transfer technology, and the economic balance between utility consumption and heat-exchanger size.

A smaller value of $\Delta T_{\min}$ generally permits greater internal heat recovery but requires closer temperature approaches and therefore larger heat-transfer areas. A larger value reduces the required exchanger area but normally increases the demand for external heating and cooling utilities.

The practical question is therefore not simply

How much heat can be recovered?

but rather

How much heat can be recovered while maintaining an acceptable minimum temperature difference throughout the process?

2.2 A Simple Four-Stream Example

The procedure will be illustrated using the four process streams introduced in Section 1. Their data are repeated in Table 2 for convenience.

Table 2: Process data for the four-stream example.

Stream	T_s (°C)	T_t (°C)	$\dot{m}$ (kg/s)	c_p (kJ/(kg · K))	CP (kW/K)
H1	220	80	1.0	2.0	2.0
H2	180	60	1.0	1.5	1.5
C1	20	180	1.0	1.5	1.5
C2	50	150	1.0	2.0	2.0

The hot streams H1 and H2 must be cooled, whereas the cold streams C1 and C2 must be heated. For the present example, the minimum allowable temperature difference is selected as

$$\Delta T_{\min} = 10 \text{ K}. \quad (10)$$

The objective is not yet to decide which hot stream should be connected to which cold stream. The immediate task is to determine the maximum overall heat recovery that is thermodynamically possible for the given stream data and the selected value of $\Delta T_{\min}$.

2.3 Temperature Shifting

Directly enforcing the minimum temperature difference between every possible combination of hot and cold streams would make the analysis unnecessarily complicated. Pinch Analysis therefore introduces a simple temperature transformation.

Each hot-stream temperature is shifted downward by half of the minimum temperature difference:

$$T_h^* = T_h - \frac{\Delta T_{\min}}{2}. \quad (11)$$

Each cold-stream temperature is shifted upward by the same amount:

$$T_c^* = T_c + \frac{\Delta T_{\min}}{2}. \quad (12)$$

The superscript * denotes a shifted temperature. For the selected value

$$\frac{\Delta T_{\min}}{2} = 5 \text{ K},$$

all hot-stream temperatures are reduced by 5 K, while all cold-stream temperatures are increased by 5 K.

For example, the hot stream H1 is shifted from

$$220^\circ\text{C} \rightarrow 80^\circ\text{C}$$

to

$$215^\circ\text{C} \rightarrow 75^\circ\text{C}.$$

The cold stream C1 is shifted from

$$20^\circ\text{C} \rightarrow 180^\circ\text{C}$$

to

$$25^\circ\text{C} \rightarrow 185^\circ\text{C}.$$

Applying the same procedure to all four streams gives the shifted temperatures in Table 3.

Table 3: Shifted temperatures for the four-stream example.

Stream	T_s^* ($^\circ\text{C}$)	T_t^* ($^\circ\text{C}$)
H1	215	75
H2	175	55
C1	25	185
C2	55	155

The reason for this transformation is fundamental. If a shifted hot temperature and a shifted cold temperature are equal, the corresponding real temperatures differ by exactly $\Delta T_{\min}$. Indeed, if

$$T_h^* = T_c^*, \quad (13)$$

then Equations (11) and (12) imply

$$T_h - T_c = \Delta T_{\min}. \quad (14)$$

The minimum temperature approach is therefore incorporated implicitly into the shifted temperature scale. The remaining analysis can be performed using these shifted temperatures without repeatedly checking the real hot-to-cold temperature difference.

2.4 Temperature Levels and Intervals

The next step is straightforward. All shifted supply and target temperatures are written into a single list. For the four-stream example, this gives

$$215, 75, 175, 55, 25, 185, 55, 155.$$

Duplicate values are then removed. The temperature 55 °C, for example, occurs as the shifted target temperature of H2 and as the shifted supply temperature of C2. It represents one common temperature level and is therefore retained only once.

The remaining shifted temperatures are sorted from highest to lowest:

$$215, 185, 175, 155, 75, 55, 25. \quad (15)$$

Adjacent temperature levels define the temperature intervals shown in Table 4.

Table 4: Shifted temperature intervals for the four-stream example.

Interval	Upper Temperature (°C)	Lower Temperature (°C)
1	215	185
2	185	175
3	175	155
4	155	75
5	75	55
6	55	25

Each interval represents one temperature layer of the process. Within a given interval, the relevant question is which hot and cold streams span the complete temperature range.

A stream is active in an interval if its shifted temperature range covers that entire interval. Once the active streams have been identified, their heat-capacity flow rates can be combined to determine the energy balance of the interval.

Removing duplicate temperatures is important because two identical temperature levels do not define a finite interval. Retaining duplicate values would create an artificial interval with zero temperature width and therefore no physical significance.

2.5 The Heat Cascade

For each temperature interval, the heat-capacity flow rates of the active hot streams are added together. The same calculation is performed for the active cold streams.

The resulting net heat-capacity flow rate is

$$CP_{\text{net}} = \sum CP_{\text{hot}} - \sum CP_{\text{cold}}. \quad (16)$$

Multiplying this value by the width of the temperature interval gives the interval heat balance:

$$\Delta H = CP_{\text{net}} \Delta T. \quad (17)$$

A positive value of ΔH represents a heat surplus, whereas a negative value represents a heat deficit.

As a numerical illustration, consider the interval from 175 °C to 155 °C. The interval width is

$$\Delta T = 175 - 155 = 20 \text{ K}. \quad (18)$$

Both hot streams are active in this interval, giving

$$\sum CP_{\text{hot}} = 2.0 + 1.5 = 3.5 \text{ kW/K}.$$

Only C1 is active over the full interval, giving

$$\sum CP_{\text{cold}} = 1.5 \text{ kW/K}.$$

The net heat-capacity flow rate is therefore

$$CP_{\text{net}} = 3.5 - 1.5 = 2.0 \text{ kW/K}. \quad (19)$$

The interval has the heat surplus

$$\Delta H = 2.0 \text{ kW/K} \cdot 20 \text{ K} = 40 \text{ kW}. \quad (20)$$

The same calculation is performed for every temperature interval. The individual interval balances are then accumulated from the highest shifted temperature to the lowest. This running accumulation is known as the *heat cascade*.

The heat cascade can be understood as a bookkeeping system for thermal energy. Heat released at a high temperature can be carried downward and used by cold streams at lower temperatures. The cascade keeps track of the available heat as it passes through the successive temperature intervals.

The first cascade is started without any external heating. If the accumulated balance remains non-negative throughout the complete temperature range, no external hot utility is required. If the balance becomes negative, the process contains a heat deficit that cannot be covered by heat released at higher temperatures.

The largest negative value encountered in this initial cascade determines the minimum amount of external hot utility that must be added at the top. Adding this amount shifts the entire cascade upward so that all residual heat balances become non-negative.

The final residual at the bottom of the adjusted cascade is the minimum external cold utility. The heat cascade therefore determines both the minimum heating requirement and the minimum cooling requirement of the process.

2.6 The Pinch Point

After the minimum hot utility has been added, the adjusted heat cascade contains no negative residual values. At least one temperature level normally has a residual of zero. This location defines the *pinch point*.

The pinch point represents the thermodynamic bottleneck of the process. At this point, no excess heat is available to pass from the higher-temperature region to the lower-temperature region.

The process is therefore divided into two distinct regions:

- Above the pinch, the process has an overall heat deficit and requires external heating.
- Below the pinch, the process has an overall heat surplus and requires external cooling.

The minimum hot and cold utility requirements obtained from the cascade are commonly denoted by

$$Q_{H,\min} \quad \text{and} \quad Q_{C,\min}. \quad (21)$$

The pinch is first identified on the shifted temperature scale. The corresponding physical hot-side and cold-side pinch temperatures differ by exactly $\Delta T_{\min}$.

The pinch point is not merely one result among many. It determines how the subsequent heat exchanger network must be organized. Transferring heat across the pinch or placing utilities on the wrong side of the pinch would increase the external heating and cooling requirements above the minimum values identified by the cascade.

In many respects, the pinch point is the single most important outcome of the entire analysis. Once it has been identified, the minimum utility targets are known and the engineer can begin designing a heat exchanger network that respects the thermodynamic limits of the process.

2.7 Complete Results for a Four-Stream Example

If the original four-stream data from Table 1 are carried through the complete pinch calculation, the resulting heat cascade remains non-negative. That example therefore represents a special threshold case with

$$Q_{H,\min} = 0 \text{ kW}, \quad (22)$$

a minimum cooling requirement of

$$Q_{C,\min} = 20 \text{ kW}, \quad (23)$$

and no internal pinch point. Such cases are possible in practice, but they are not ideal for illustrating the usual pinch procedure.

For this reason, the present subsection uses the modified four-stream example given in Table 5. The example has been selected so that the adjusted heat cascade reaches zero at an internal temperature level and therefore exhibits a genuine internal pinch point.

Table 5: Process data for the modified four-stream example.

Stream	T_s (°C)	T_t (°C)	$\dot{m}$ (kg/s)	c_p (kJ/(kg · K))	CP (kW/K)
H1	240	100	1.0	3.5	3.5
H2	180	120	1.0	2.5	2.5
C1	100	220	1.0	4.0	4.0
C2	120	160	1.0	3.5	3.5

For the selected minimum temperature difference

$$\Delta T_{\min} = 10 \text{ K}, \quad (24)$$

the hot-stream temperatures are shifted downward by 5 K, while the cold-stream temperatures are shifted upward by 5 K. The resulting shifted temperatures are shown in Table 6.

Table 6: Shifted temperatures for the modified four-stream example.

Stream	T_s^* (°C)	T_t^* (°C)
H1	235	95
H2	175	115
C1	105	225
C2	125	165

Collecting the shifted supply and target temperatures, removing duplicates, and sorting the remaining values from highest to lowest gives

$$235, 225, 175, 165, 125, 115, 105, 95. \quad (25)$$

These eight temperature levels define seven temperature intervals. The complete interval calculation and heat cascade are shown in Table 7.

Table 7: Complete heat cascade for the modified four-stream example.

Interval	Upper T^* (°C)	Lower T^* (°C)	ΔT (K)	$\sum CP_H$ (kW/K)	$\sum CP_C$ (kW/K)	CP_{net} (kW/K)	ΔH (kW)	Unadjusted (kW)	Adjusted (kW)
1	235	225	10	3.5	0.0	3.5	35	35	65
2	225	175	50	3.5	4.0	-0.5	-25	10	40
3	175	165	10	6.0	4.0	2.0	20	30	60
4	165	125	40	6.0	7.5	-1.5	-60	-30	0
5	125	115	10	6.0	4.0	2.0	20	-10	20
6	115	105	10	3.5	4.0	-0.5	-5	-15	15
7	105	95	10	3.5	0.0	3.5	35	20	50

The first interval extends from 235 °C to 225 °C. Only H1 is active in this interval. The net heat-capacity flow rate is

$$CP_{\text{net},1} = 3.5 \text{ kW/K}, \quad (26)$$

and the corresponding interval heat surplus is

$$\Delta H_1 = 3.5 \text{ kW/K} \cdot 10 \text{ K} = 35 \text{ kW}. \quad (27)$$

In the second interval, H1 and C1 are active. Consequently,

$$CP_{\text{net},2} = 3.5 - 4.0 \quad (28)$$

$$= -0.5 \text{ kW/K}, \quad (29)$$

and

$$\Delta H_2 = -0.5 \text{ kW/K} \cdot 50 \text{ K} \quad (30)$$

$$= -25 \text{ kW}. \quad (31)$$

In the third interval, both hot streams are active, while only C1 is active. The interval balance is therefore

$$\Delta H_3 = (6.0 - 4.0) \text{ kW/K} \cdot 10 \text{ K} \quad (32)$$

$$= 20 \text{ kW}. \quad (33)$$

The fourth interval extends from 165 °C to 125 °C. Both hot streams and both cold streams are active. Hence,

$$CP_{\text{net},4} = 6.0 - 7.5 \quad (34)$$

$$= -1.5 \text{ kW/K}, \quad (35)$$

and the interval heat deficit is

$$\Delta H_4 = -1.5 \text{ kW/K} \cdot 40 \text{ K} \quad (36)$$

$$= -60 \text{ kW}. \quad (37)$$

Starting the initial heat cascade with zero external heating produces the cumulative values

$$35, 10, 30, -30, -10, -15, 20 \text{ kW}. \quad (38)$$

The lowest value of the unadjusted cascade is

$$-30 \text{ kW}. \quad (39)$$

The minimum external hot utility is therefore

$$\boxed{Q_{H,\min} = 30 \text{ kW}}. \quad (40)$$

Adding 30 kW at the top shifts the complete cascade upward. The adjusted cascade becomes

$$65, 40, 60, 0, 20, 15, 50 \text{ kW}. \quad (41)$$

The final residual at the lowest shifted temperature level is

$$\boxed{Q_{C,\min} = 50 \text{ kW}}. \quad (42)$$

The adjusted cascade reaches zero after the fourth temperature interval. The corresponding shifted pinch temperature is therefore

$$\boxed{T_{\text{pinch}}^* = 125^\circ \text{C}}. \quad (43)$$

Reversing the temperature shift gives the physical hot-side pinch temperature

$$T_{H,\text{pinch}} = T_{\text{pinch}}^* + \frac{\Delta T_{\min}}{2} \quad (44)$$

$$= 125 + 5 \quad (45)$$

$$= \boxed{130^\circ \text{C}}, \quad (46)$$

and the physical cold-side pinch temperature

$$T_{C,\text{pinch}} = T_{\text{pinch}}^* - \frac{\Delta T_{\min}}{2} \quad (47)$$

$$= 125 - 5 \quad (48)$$

$$= \boxed{120^\circ \text{C}}. \quad (49)$$

The required minimum temperature difference is therefore recovered exactly:

$$T_{H,\text{pinch}} - T_{C,\text{pinch}} = 130 - 120 = 10 \text{ K} = \Delta T_{\min}. \quad (50)$$

The utility targets can be checked independently using the overall process heat balance. The total heat released by the hot streams is

$$Q_{\text{hot}} = 3.5(240 - 100) + 2.5(180 - 120) \quad (51)$$

$$= 490 + 150 \quad (52)$$

$$= 640 \text{ kW}. \quad (53)$$

The total heat required by the cold streams is

$$Q_{\text{cold}} = 4.0(220 - 100) + 3.5(160 - 120) \quad (54)$$

$$= 480 + 140 \quad (55)$$

$$= 620 \text{ kW}. \quad (56)$$

The complete process energy balance is therefore

$$Q_{H,\text{min}} + Q_{\text{hot}} = 30 \text{ kW} + 640 \text{ kW} \quad (57)$$

$$= 670 \text{ kW}, \quad (58)$$

while

$$Q_{\text{cold}} + Q_{C,\text{min}} = 620 \text{ kW} + 50 \text{ kW} \quad (59)$$

$$= 670 \text{ kW}. \quad (60)$$

Both sides of the energy balance agree exactly.

The principal results of the modified four-stream example are summarized in Table 8.

Table 8: Thermodynamic targets for the modified four-stream example.

Quantity	Result
Minimum hot utility $Q_{H,\text{min}}$	30 kW
Minimum cold utility $Q_{C,\text{min}}$	50 kW
Shifted pinch temperature T_{pinch}^*	125 °C
Hot-side pinch temperature $T_{H,\text{pinch}}$	130 °C
Cold-side pinch temperature $T_{C,\text{pinch}}$	120 °C

The pinch divides the process into a heat-deficit region above the pinch and a heat-surplus region below the pinch. These two regions must be treated separately when the heat exchanger network is constructed.

The thermodynamic targeting stage is now complete. The minimum utility requirements and the physical pinch temperatures are known.

2.8 From Thermodynamic Targets to Practical Implementation

The pinch analysis determines the thermodynamic limits of heat recovery. In particular, it provides the minimum utility targets and identifies the pinch point. These results describe what is thermodynamically achievable for a given set of process streams and a specified minimum temperature difference.

However, the existence of a thermodynamically feasible solution does not necessarily imply that the same solution can be implemented easily in a real industrial plant. At this point it is useful to distinguish between three different situations.

Retrofit Projects. In a retrofit project, an existing process plant is analysed and improved. The plant layout, piping routes, equipment locations, pressure levels, safety requirements, and operating constraints are already fixed.

Although the pinch analysis may identify attractive heat recovery opportunities, implementing them may require substantial modifications to an existing facility. Two streams may be thermodynamically well matched while being located in completely different parts of the plant. Additional piping, structural modifications, pressure losses, safety considerations, or downtime requirements may significantly reduce the practical attractiveness of a theoretically optimal solution.

Consequently, retrofit projects often seek a compromise between the theoretical utility targets and the cost of modifying the existing plant.

Grassroots Design. In a grassroots design project, the process is designed from the ground up. Because the process layout has not yet been fixed, the results of the pinch analysis can influence the arrangement of equipment, piping, and utilities from the earliest stages of design.

In this situation, the process can be configured so that thermally compatible streams are placed in suitable locations and the Heat Exchanger Network is incorporated directly into the overall process design. As a result, a significantly larger fraction of the theoretical heat-recovery potential can often be realized than in a retrofit project.

Intermediate Cases. Many industrial projects lie between these two extremes. Existing plants are expanded, modified, or partially rebuilt while retaining substantial portions of the original equipment. Such projects are often referred to as revamps, brownfield developments, or plant modernization projects.

In these situations, some design freedom exists, but practical constraints remain important. The pinch targets provide valuable guidance, but engineering judgement is required to balance thermal performance, capital investment, operability, and implementation cost.

The distinction between thermodynamic targets and practical implementation is therefore important. Pinch Analysis identifies the best theoretically achievable utility targets. The subsequent Heat Exchanger Network design determines how closely these targets can be approached within the constraints of a real engineering project.

This transition from thermodynamic targeting to actual network design is the subject of the next section.

3 From Pinch Analysis to Heat Exchanger Network Design

The pinch analysis performed in the previous section determines the thermodynamic targets of the process. At this point the engineer knows

$$Q_{H,\min}, \quad Q_{C,\min}, \quad T_{\text{pinch}},$$

and therefore has a clear understanding of the best possible heat recovery that can be achieved while respecting the prescribed minimum temperature difference.

However, the pinch analysis itself does not specify how heat should be transferred between individual streams. The next step is therefore to construct a Heat Exchanger Network (HEN) that approaches the previously identified thermodynamic targets.

3.1 The Role of the Pinch Point

The pinch point is not merely a numerical result of the heat cascade. It acts as the organizing principle for the entire network design.

Once the pinch point has been established, the process is divided into two thermodynamically distinct regions. These regions must be treated independently during network synthesis. Violating this separation leads to avoidable increases in utility consumption and moves the design away from the thermodynamic targets identified by the pinch analysis.

Consequently, the pinch point functions as the central reference point for all subsequent network-design decisions.

3.2 The Two Network Regions

The pinch divides the process into two regions.

The region *above the pinch* corresponds to higher temperatures. From an overall energy perspective, this region exhibits a heat deficit. Additional heat must ultimately be supplied through external hot utilities.

The region *below the pinch* corresponds to lower temperatures. This region exhibits a heat surplus. Excess heat must ultimately be removed through external cooling utilities.

The distinction is illustrated conceptually as

Above the Pinch $\longrightarrow$ Heat Deficit,

Below the Pinch $\longrightarrow$ Heat Surplus.

Any network design must respect this thermodynamic separation.

It should be emphasized that this classification refers to the overall energy balance of the region and not to the local behavior of individual streams. Hot streams may still exist above the pinch and cold streams may still exist below the pinch. The distinction concerns the net heat balance of the region rather than the classification of individual streams.

3.3 The Three Fundamental Pinch Rules

Three classical design rules follow directly from the pinch concept.

1. No heat should be transferred across the pinch.
2. No cooling should be applied above the pinch.
3. No heating should be applied below the pinch.

The first rule prevents heat from being transported from the heat-surplus region into the heat-deficit region in a manner that increases utility consumption.

The second rule prevents valuable heat from being discarded in a region that already lacks sufficient thermal energy.

The third rule prevents additional heat from being introduced into a region that already contains more heat than required.

Violating any of these rules increases both operating costs and utility requirements. Consequently, they form the foundation of pinch-based network design.

3.4 Stream Matching and Heat-Capacity Flow Rates

Once the process has been divided at the pinch, suitable hot and cold streams must be paired.

At this stage, the heat-capacity flow rate

$$CP = \dot{m} c_p$$

becomes particularly important.

Suppose heat is transferred between a hot stream and a cold stream. A stream with a large value of CP undergoes only a small temperature change for a given heat duty, whereas a stream with a small value of CP experiences a larger temperature change.

For this reason, practical matching rules are commonly applied.

Above the pinch, feasible matches generally satisfy

$$CP_{\text{cold}} \geq CP_{\text{hot}}. \quad (61)$$

The intuition is straightforward. The cold stream should be thermally strong enough to absorb the transferred heat without its temperature rising excessively quickly.

Below the pinch, the corresponding guideline becomes

$$CP_{\text{hot}} \geq CP_{\text{cold}}. \quad (62)$$

Here the hot stream should be thermally strong enough to release heat without its temperature dropping excessively quickly.

The purpose of these rules is to reduce the likelihood of temperature crossing and to maintain a feasible temperature driving force throughout the exchanger.

These criteria are best viewed as practical design guidelines for selecting among multiple feasible stream matches. They should not be interpreted as universal mathematical constraints that apply to every possible heat exchanger configuration. Their primary purpose is to help identify matches that are less likely to create temperature-crossing problems during network synthesis.

Example Above the Pinch. Consider a hot stream and a cold stream with

$$CP_{\text{hot}} = 2 \text{ kW/K}$$

and

$$CP_{\text{cold}} = 4 \text{ kW/K}.$$

Suppose a heat duty of

$$Q = 40 \text{ kW}$$

is transferred between the two streams.

The temperature change of the hot stream is

$$\Delta T_{\text{hot}} = \frac{Q}{CP_{\text{hot}}} = \frac{40}{2} = 20 \text{ K}.$$

The temperature change of the cold stream is

$$\Delta T_{\text{cold}} = \frac{Q}{CP_{\text{cold}}} = \frac{40}{4} = 10 \text{ K}.$$

The cold stream warms up more slowly than the hot stream cools down. Consequently, the temperature difference tends to widen as one moves away from the pinch. This behavior is generally favorable and is consistent with

$$CP_{\text{cold}} \geq CP_{\text{hot}}.$$

Counterexample Above the Pinch. Now consider

$$CP_{\text{hot}} = 4 \text{ kW/K}$$

and

$$CP_{\text{cold}} = 2 \text{ kW/K}.$$

For the same heat duty of

$$Q = 40 \text{ kW},$$

the temperature changes become

$$\Delta T_{\text{hot}} = \frac{40}{4} = 10 \text{ K}$$

and

$$\Delta T_{\text{cold}} = \frac{40}{2} = 20 \text{ K}.$$

The cold stream now heats up twice as fast as the hot stream cools down. The temperature difference shrinks as one moves away from the pinch and may eventually vanish. A temperature crossover can occur, making the exchanger physically infeasible.

This is precisely the situation that the guideline

$$CP_{\text{cold}} \geq CP_{\text{hot}}$$

seeks to avoid.

Example Below the Pinch. Below the pinch, the logic is reversed.

Consider

$$CP_{\text{hot}} = 4 \text{ kW/K}$$

and

$$CP_{\text{cold}} = 2 \text{ kW/K}.$$

For a transferred heat load of

$$Q = 40 \text{ kW},$$

the hot stream experiences a temperature change of

$$\Delta T_{\text{hot}} = 10 \text{ K},$$

while the cold stream experiences

$$\Delta T_{\text{cold}} = 20 \text{ K}.$$

In this case, the hot stream remains thermally stronger than the cold stream, which is consistent with

$$CP_{\text{hot}} \geq CP_{\text{cold}}.$$

As above, the rule is intended to reduce the likelihood of temperature crossing and to facilitate feasible network construction.

3.5 The Single-Match Case

The heat-capacity-flow rules are particularly important when several possible stream matches exist. In such situations the engineer must choose between competing alternatives.

A special case arises when only one hot stream and one cold stream are available in a region.

In this situation no partner selection is possible. If internal heat recovery is desired, the two streams must exchange heat with each other. The design problem therefore shifts from partner selection to the determination of an appropriate heat duty.

The governing design variable then becomes the exchanger heat duty rather than the choice of partner stream. By transferring only a suitable fraction of the available heat, the minimum temperature difference can still be maintained and temperature crossing can be avoided.

Thus, in a one-to-one situation the pairing itself is mandatory, whereas the amount of heat exchanged remains adjustable.

3.6 Stream Splitting

In some situations the heat-capacity-flow rules cannot be satisfied by the available unsplit streams. A common solution is stream splitting.

The idea is simple. A process stream is divided into two or more branches, each carrying only part of the original mass flow rate.

Because

$$CP = \dot{m} c_p,$$

splitting a stream also divides its heat-capacity flow rate.

For example, a stream with

$$CP = 8 \text{ kW/K}$$

may be divided into two equal branches:

$$8 = 4 + 4.$$

Alternatively, unequal branches may be created:

$$8 = 5 + 3.$$

The total thermal capacity of the original stream is preserved, but the individual branches can now participate in different stream matches.

Stream splitting therefore creates additional design flexibility and often allows feasible matches to be constructed when the original unsplit stream would violate the heat-capacity-flow criteria.

It should be noted that stream splitting does not create additional heat recovery potential by itself. The thermodynamic targets remain unchanged. Instead, stream splitting provides additional flexibility for constructing a network that approaches the utility targets and pinch constraints identified by the pinch analysis.

In practice, stream splitting is one of the most important tools available to the network designer.

3.7 From the Grid Diagram to a Complete Network

The first graphical representation of a pinch-based design is normally a grid diagram.

The grid diagram displays:

- the hot streams,
- the cold streams,
- the pinch location,
- the regions above and below the pinch,
- and the connections established between individual streams.

The network is typically constructed beginning at the pinch and then expanded outward. Internal heat recovery opportunities are exhausted as far as possible before external heaters or coolers are introduced.

Once suitable stream matches have been identified, individual heat exchangers can be assigned, heat duties can be calculated, and utility requirements can be located at the appropriate boundaries of the network.

For large industrial systems this process is often supported by specialized software. Nevertheless, the underlying logic remains the same: the pinch analysis determines the thermodynamic targets, while the network design seeks to achieve those targets using an actual set of heat exchangers and utilities.

Two Levels of Presentation. The following worked example is presented in two complementary forms. The first presentation provides a detailed, step-by-step derivation of the pinch-compliant network. The second summarizes the same solution in a more compact engineering format. Both presentations describe the same network and lead to identical utility targets and thermal balances.

3.8 Worked Example: Detailed Development of a Feasible Network

The pinch analysis in Subsection 2.7 established the thermodynamic targets for the modified four-stream example. The relevant results are

$$Q_{H,\min} = 30 \text{ kW}, \quad Q_{C,\min} = 50 \text{ kW}, \quad (63)$$

together with the physical pinch temperatures

$$T_{H,\text{pinch}} = 130^\circ\text{C}, \quad T_{C,\text{pinch}} = 120^\circ\text{C}. \quad (64)$$

The thermodynamic targets are therefore known. The remaining task is to construct a heat exchanger network that satisfies all process-stream targets, respects the minimum temperature difference, avoids heat transfer across the pinch, and reproduces the minimum utility requirements in Equation (63).

3.8.1 Stream Segments Above and Below the Pinch

The first step is to divide every stream that crosses the pinch into an upper and a lower segment. This division is a thermodynamic separation at the pinch and should not be confused with a physical split into parallel branches.

Above the pinch, the hot-stream duties are

$$\begin{aligned} Q_{H1}^{\text{above}} &= CP_{H1} (T_{s,H1} - T_{H,\text{pinch}}) \\ &= 3.5(240 - 130) \\ &= 385 \text{ kW}, \end{aligned} \quad (65)$$

and

$$\begin{aligned}
Q_{H2}^{\text{above}} &= CP_{H2} (T_{s,H2} - T_{H,\text{pinch}}) \\
&= 2.5(180 - 130) \\
&= 125 \text{ kW}.
\end{aligned} \tag{66}$$

The corresponding cold-stream duties above the pinch are

$$\begin{aligned}
Q_{C1}^{\text{above}} &= CP_{C1} (T_{t,C1} - T_{C,\text{pinch}}) \\
&= 4.0(220 - 120) \\
&= 400 \text{ kW},
\end{aligned} \tag{67}$$

and

$$\begin{aligned}
Q_{C2}^{\text{above}} &= CP_{C2} (T_{t,C2} - T_{C,\text{pinch}}) \\
&= 3.5(160 - 120) \\
&= 140 \text{ kW}.
\end{aligned} \tag{68}$$

The total hot-process duty available above the pinch is therefore

$$Q_H^{\text{above}} = 385 + 125 = 510 \text{ kW}, \tag{69}$$

whereas the cold streams require

$$Q_C^{\text{above}} = 400 + 140 = 540 \text{ kW}. \tag{70}$$

The difference is

$$Q_C^{\text{above}} - Q_H^{\text{above}} = 540 - 510 = 30 \text{ kW}, \tag{71}$$

which agrees with $Q_{H,\text{min}}$.

Below the pinch, the available hot-stream duties are

$$Q_{H1}^{\text{below}} = 3.5(130 - 100) = 105 \text{ kW}, \tag{72}$$

$$Q_{H2}^{\text{below}} = 2.5(130 - 120) = 25 \text{ kW}. \tag{73}$$

Only C1 extends below the pinch. Its lower duty is

$$Q_{C1}^{\text{below}} = 4.0(120 - 100) = 80 \text{ kW}. \tag{74}$$

The lower region therefore contains the net heat surplus

$$Q_H^{\text{below}} - Q_C^{\text{below}} = (105 + 25) - 80 = 50 \text{ kW}, \tag{75}$$

which agrees with $Q_{C,\text{min}}$.

Table 9 summarizes these stream segments.

Table 9: Stream duties above and below the pinch.

Stream	CP (kW/K)	Duty Above Pinch (kW)	Duty Below Pinch (kW)
H1	3.5	385	105
H2	2.5	125	25
C1	4.0	400	80
C2	3.5	140	0

3.8.2 Network Construction Above the Pinch

Above the pinch, the preferred matching guideline is

$$CP_{\text{cold}} \geq CP_{\text{hot}}. \quad (76)$$

A feasible upper network can be constructed using three internal heat exchangers.

Exchanger E_1 : H1 with C2. The complete upper duty of C2 is supplied by H1:

$$Q_{E_1} = 140 \text{ kW}. \quad (77)$$

The cold-stream temperature change is

$$\Delta T_{C2} = \frac{140}{3.5} = 40 \text{ K}, \quad (78)$$

so that C2 is heated from

$$120^\circ\text{C} \quad \text{to} \quad 160^\circ\text{C}.$$

The corresponding H1 segment is cooled from

$$170^\circ\text{C} \quad \text{to} \quad 130^\circ\text{C}.$$

The terminal temperature differences are

$$\Delta T_{\text{hot end}} = 170 - 160 = 10 \text{ K}, \quad (79)$$

$$\Delta T_{\text{cold end}} = 130 - 120 = 10 \text{ K}. \quad (80)$$

Thus, E_1 exactly satisfies the selected minimum temperature difference at both ends. Furthermore,

$$CP_{C2} = CP_{H1} = 3.5 \text{ kW/K}. \quad (81)$$

Exchanger E_2 : H2 with C1. The complete upper duty of H2 is transferred to C1:

$$Q_{E_2} = 125 \text{ kW}. \quad (82)$$

H2 is cooled from

$$180^\circ\text{C} \quad \text{to} \quad 130^\circ\text{C}.$$

C1 is heated from the cold-side pinch temperature to

$$T_{C1,\text{out},E_2} = 120 + \frac{125}{4.0} = 151.25^\circ\text{C}. \quad (83)$$

The terminal temperature differences are

$$180 - 151.25 = 28.75 \text{ K}, \quad (84)$$

$$130 - 120 = 10 \text{ K}. \quad (85)$$

The relevant heat-capacity-flow condition is also satisfied:

$$CP_{C1} = 4.0 \geq 2.5 = CP_{H2}. \quad (86)$$

Exchanger E_3 : H1 with C1. After supplying C2, the remaining upper duty of H1 is

$$Q_{E_3} = 385 - 140 = 245 \text{ kW}. \quad (87)$$

H1 is cooled from

$$240^\circ\text{C} \quad \text{to} \quad 170^\circ\text{C},$$

because

$$240 - \frac{245}{3.5} = 170^\circ\text{C}. \quad (88)$$

C1 enters E_3 at 151.25°C and leaves at

$$T_{C1,\text{out},E_3} = 151.25 + \frac{245}{4.0} = 212.5^\circ\text{C}. \quad (89)$$

The terminal temperature differences are

$$240 - 212.5 = 27.5 \text{ K}, \quad (90)$$

$$170 - 151.25 = 18.75 \text{ K}. \quad (91)$$

The match therefore respects both the minimum temperature difference and

$$CP_{C1} = 4.0 \geq 3.5 = CP_{H1}. \quad (92)$$

Hot Utility. After E_2 and E_3 , C1 has reached 212.5°C . Its remaining heating requirement is

$$Q_{\text{heater}} = CP_{C1}(220 - 212.5) \quad (93)$$

$$= 4.0(7.5) \quad (94)$$

$$= 30 \text{ kW}. \quad (95)$$

Hence,

$$Q_{\text{heater}} = Q_{H,\text{min}}. \quad (96)$$

No cooler is used above the pinch.

3.8.3 Stream Splitting and Network Construction Below the Pinch

Below the pinch, the preferred matching guideline is

$$CP_{\text{hot}} \geq CP_{\text{cold}}. \quad (97)$$

Only C1 requires heat below the pinch, and

$$CP_{C1} = 4.0 \text{ kW/K}. \quad (98)$$

This value is larger than both available hot-stream values:

$$CP_{H1} = 3.5 \text{ kW/K}, \quad CP_{H2} = 2.5 \text{ kW/K}. \quad (99)$$

The preferred matching condition cannot therefore be satisfied by the unsplit cold stream at the pinch. C1 is consequently divided into two parallel branches:

$$CP_{C1} = CP_{C1A} + CP_{C1B}, \quad (100)$$

with

$$4.0 = 2.75 + 1.25. \quad (101)$$

The corresponding mass-flow fractions are

$$f_A = \frac{2.75}{4.0} = 0.6875, \quad (102)$$

$$f_B = \frac{1.25}{4.0} = 0.3125. \quad (103)$$

Thus, branch C1A carries 68.75% of the original C1 mass flow, whereas branch C1B carries 31.25%. Both branches are heated from 100°C to the cold-side pinch temperature of 120°C .

Exchanger E_4 : H1 with C1A. The required duty for branch C1A is

$$Q_{E_4} = CP_{C1A}(120 - 100) \quad (104)$$

$$= 2.75(20) \quad (105)$$

$$= 55 \text{ kW}. \quad (106)$$

H1 enters E_4 at 130°C and leaves at

$$T_{H1,\text{out},E_4} = 130 - \frac{55}{3.5} \quad (107)$$

$$= 114.29^\circ\text{C}. \quad (108)$$

The terminal temperature differences are

$$130 - 120 = 10 \text{ K}, \quad (109)$$

$$114.29 - 100 = 14.29 \text{ K}. \quad (110)$$

The heat-capacity-flow condition is satisfied:

$$CP_{H1} = 3.5 \geq 2.75 = CP_{C1A}. \quad (111)$$

Exchanger E_5 : H2 with C1B. The required duty for branch C1B is

$$Q_{E_5} = CP_{C1B}(120 - 100) \quad (112)$$

$$= 1.25(20) \quad (113)$$

$$= 25 \text{ kW}. \quad (114)$$

This duty cools H2 from

$$130^\circ\text{C} \quad \text{to} \quad 120^\circ\text{C},$$

because

$$130 - \frac{25}{2.5} = 120^\circ\text{C}. \quad (115)$$

The terminal temperature differences are

$$130 - 120 = 10 \text{ K}, \quad (116)$$

$$120 - 100 = 20 \text{ K}. \quad (117)$$

The heat-capacity-flow condition is again satisfied:

$$CP_{H2} = 2.5 \geq 1.25 = CP_{C1B}. \quad (118)$$

Cold Utility. The lower segment of H1 contains

$$Q_{H1}^{\text{below}} = 105 \text{ kW}.$$

After transferring 55 kW in E_4 , the remaining heat is

$$Q_{\text{cooler}} = 105 - 55 = 50 \text{ kW}. \quad (119)$$

The cooler reduces the H1 temperature from approximately 114.29 °C to its target temperature of 100 °C. Thus,

$$Q_{\text{cooler}} = Q_{C,\text{min}}. \quad (120)$$

No heater is used below the pinch.

3.8.4 Schematic Grid Diagram

Figure 2 shows a schematic grid representation of the resulting network. The diagram is not drawn to temperature scale. Its purpose is to show the stream connections, utility placement, pinch separation, and the splitting of C1 below the pinch.

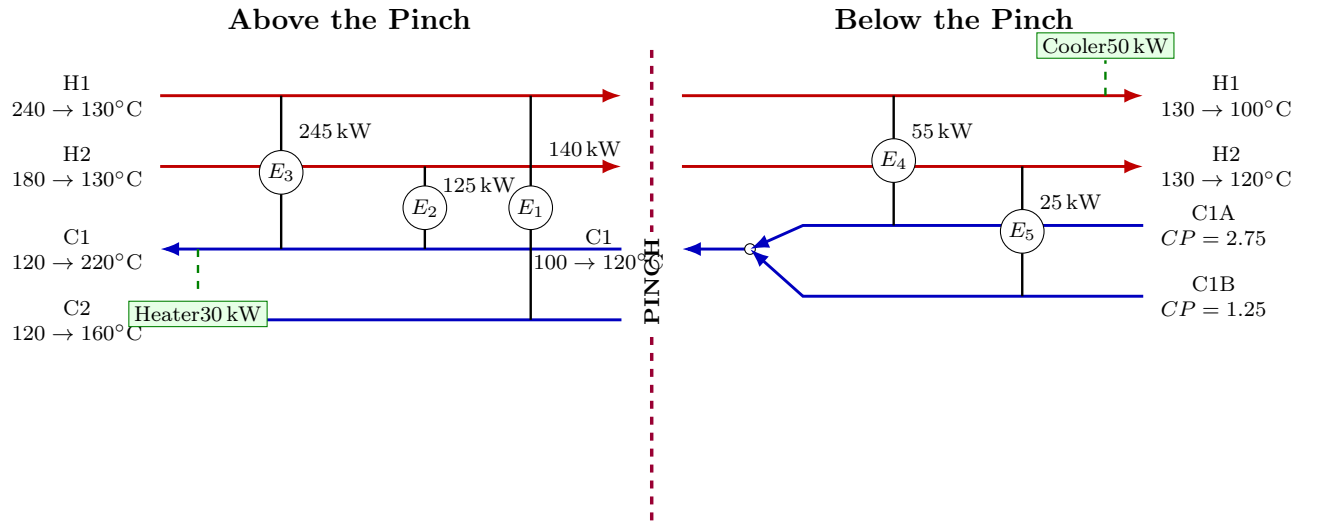


Figure 1: Schematic pinch-based grid diagram for the worked example. The stream directions indicate process flow and the exchanger positions are schematic rather than temperature-scaled.

3.8.5 Summary and Verification

The complete network is summarized in Table 10.

The total internal heat recovery is

$$Q_{\text{recovered}} = Q_{E_1} + Q_{E_2} + Q_{E_3} + Q_{E_4} + Q_{E_5} \quad (121)$$

$$= 140 + 125 + 245 + 55 + 25 \quad (122)$$

$$= 590 \text{ kW}. \quad (123)$$

Table 10: Heat exchangers and utilities in the worked example.

Unit	Region	Hot Side	Cold Side	Duty (kW)
E_1	Above	H1	C2	140
E_2	Above	H2	C1	125
E_3	Above	H1	C1	245
Heater	Above	Hot utility	C1	30
E_4	Below	H1	C1A	55
E_5	Below	H2	C1B	25
Cooler	Below	H1	Cold utility	50

The result can be verified independently from the cold-stream side:

$$Q_{\text{cold}} - Q_{H,\min} = 620 - 30 \quad (124)$$

$$= 590 \text{ kW}. \quad (125)$$

The same result follows from the hot-stream side:

$$Q_{\text{hot}} - Q_{C,\min} = 640 - 50 \quad (126)$$

$$= 590 \text{ kW}. \quad (127)$$

The complete energy balance is therefore

$$Q_{H,\min} + Q_{\text{hot}} = Q_{\text{cold}} + Q_{C,\min}, \quad (128)$$

or numerically,

$$30 + 640 = 620 + 50 = 670 \text{ kW}. \quad (129)$$

The network also satisfies the three fundamental pinch rules:

- no heat is transferred across the pinch,
- no cooling utility is used above the pinch,
- no heating utility is used below the pinch.

All five internal heat exchangers maintain at least the prescribed minimum temperature difference of

$$\Delta T_{\min} = 10 \text{ K}.$$

The example therefore represents a complete and feasible pinch-compliant network for the selected process data. It also illustrates why stream splitting can become necessary during network construction even when the thermodynamic targets and pinch temperatures are already known.

3.9 Worked Example: Condensed Engineering Summary

This subsection continues the modified four-stream example introduced in Subsection 2.7. The pinch analysis yielded

$$Q_{H,\min} = 30 \text{ kW}, \quad Q_{C,\min} = 50 \text{ kW}, \quad (130)$$

and the physical pinch temperatures

$$T_{H,\text{pinch}} = 130^\circ\text{C}, \quad T_{C,\text{pinch}} = 120^\circ\text{C}. \quad (131)$$

The objective is to construct one feasible network that attains these utility targets while satisfying the pinch constraints and the selected minimum temperature difference

$$\Delta T_{\min} = 10 \text{ K}. \quad (132)$$

Assumptions. The example assumes constant heat-capacity flow rates, sensible heat exchange, negligible heat losses, no pressure effects, and counter-current heat exchangers. Under these assumptions, the stream temperatures vary linearly with transferred heat. Feasibility can therefore be checked from the terminal temperature differences of each exchanger.

3.9.1 Separation at the Pinch

Streams crossing the pinch are divided into thermodynamic segments above and below the pinch. This separation is not a physical stream split. It is a bookkeeping step that prevents heat transfer across the pinch.

For a hot stream, the duties above and below the pinch are calculated from

$$Q_H^{\text{above}} = CP_H (T_s - T_{H,\text{pinch}}), \quad (133)$$

$$Q_H^{\text{below}} = CP_H (T_{H,\text{pinch}} - T_t), \quad (134)$$

provided that the stream spans the corresponding region. For a cold stream, the analogous expressions are

$$Q_C^{\text{above}} = CP_C (T_t - T_{C,\text{pinch}}), \quad (135)$$

$$Q_C^{\text{below}} = CP_C (T_{C,\text{pinch}} - T_s). \quad (136)$$

The resulting segment duties are summarized in Table 11.

Above the pinch, the hot streams provide

$$385 + 125 = 510 \text{ kW}, \quad (137)$$

whereas the cold streams require

Table 11: Stream duties above and below the pinch.

Stream	CP (kW/K)	Q^{above} (kW)	Q^{below} (kW)
H1	3.5	385	105
H2	2.5	125	25
C1	4.0	400	80
C2	3.5	140	0

$$400 + 140 = 540 \text{ kW}. \quad (138)$$

The resulting deficit is

$$540 - 510 = 30 \text{ kW} = Q_{H,\min}. \quad (139)$$

Below the pinch, the hot streams provide

$$105 + 25 = 130 \text{ kW}, \quad (140)$$

while C1 requires

$$80 \text{ kW}. \quad (141)$$

The resulting surplus is

$$130 - 80 = 50 \text{ kW} = Q_{C,\min}. \quad (142)$$

The regional balances therefore reproduce the utility targets obtained from the heat cascade.

3.9.2 Network Above the Pinch

Above the pinch, the preferred matching condition is

$$CP_{\text{cold}} \geq CP_{\text{hot}}. \quad (143)$$

A feasible network is obtained with three process-to-process exchangers and one heater. The exchanger duties and terminal temperatures are listed in Table 12.

Table 12: Network above the pinch.

Unit	Hot	Cold	Q (kW)	$T_{h,\text{in}}$ (°C)	$T_{h,\text{out}}$ (°C)	$T_{c,\text{in}}$ (°C)	$T_{c,\text{out}}$ (°C)
E_1	H1	C2	140	170	130	120	160
E_2	H2	C1	125	180	130	120	151.25
E_3	H1	C1	245	240	170	151.25	212.50
Heater	–	C1	30	–	–	212.50	220

The terminal temperature differences for the three internal exchangers are

$$E_1 : \quad 170 - 160 = 130 - 120 = 10 \text{ K}, \quad (144)$$

$$E_2 : \quad 180 - 151.25 = 28.75 \text{ K}, \quad 130 - 120 = 10 \text{ K}, \quad (145)$$

$$E_3 : \quad 240 - 212.50 = 27.50 \text{ K}, \quad 170 - 151.25 = 18.75 \text{ K}. \quad (146)$$

All terminal differences are at least $\Delta T_{\min}$. The hot utility requirement is

$$Q_{\text{heater}} = 4.0(220 - 212.5) = 30 \text{ kW} = Q_{H,\min}. \quad (147)$$

No cooling utility is used above the pinch.

3.9.3 Network Below the Pinch

Below the pinch, the preferred matching condition is

$$CP_{\text{hot}} \geq CP_{\text{cold}}. \quad (148)$$

The unsplit cold stream C1 has

$$CP_{C1} = 4.0 \text{ kW/K}, \quad (149)$$

which exceeds both available hot-stream heat-capacity flow rates. To obtain suitable matches at the pinch, C1 is divided into two parallel branches:

$$CP_{C1} = CP_{C1A} + CP_{C1B} = 2.75 + 1.25 = 4.0 \text{ kW/K}. \quad (150)$$

The corresponding flow fractions are

$$f_A = \frac{2.75}{4.0} = 0.6875, \quad f_B = \frac{1.25}{4.0} = 0.3125. \quad (151)$$

Both branches are heated from 100°C to 120°C . The resulting lower network is summarized in Table 13.

Table 13: Network below the pinch.

Unit	Hot	Cold	Q (kW)	$T_{h,\text{in}}$ ($^\circ\text{C}$)	$T_{h,\text{out}}$ ($^\circ\text{C}$)	$T_{c,\text{in}}$ ($^\circ\text{C}$)	$T_{c,\text{out}}$ ($^\circ\text{C}$)
E_4	H1	C1A	55	130	114.29	100	120
E_5	H2	C1B	25	130	120.00	100	120
Cooler	H1	—	50	114.29	100.00	—	—

The corresponding terminal temperature differences are

$$E_4 : \quad 130 - 120 = 10 \text{ K}, \quad 114.29 - 100 = 14.29 \text{ K}, \quad (152)$$

$$E_5 : \quad 130 - 120 = 10 \text{ K}, \quad 120 - 100 = 20 \text{ K}. \quad (153)$$

The remaining heat of H1 is rejected through the cooler:

$$Q_{\text{cooler}} = 105 - 55 = 50 \text{ kW} = Q_{C,\min}. \quad (154)$$

No heating utility is used below the pinch.

3.9.4 Grid Representation and Verification

The resulting topology is shown schematically in Figure 2. The exchanger locations are schematic and are not plotted on a quantitative temperature axis.

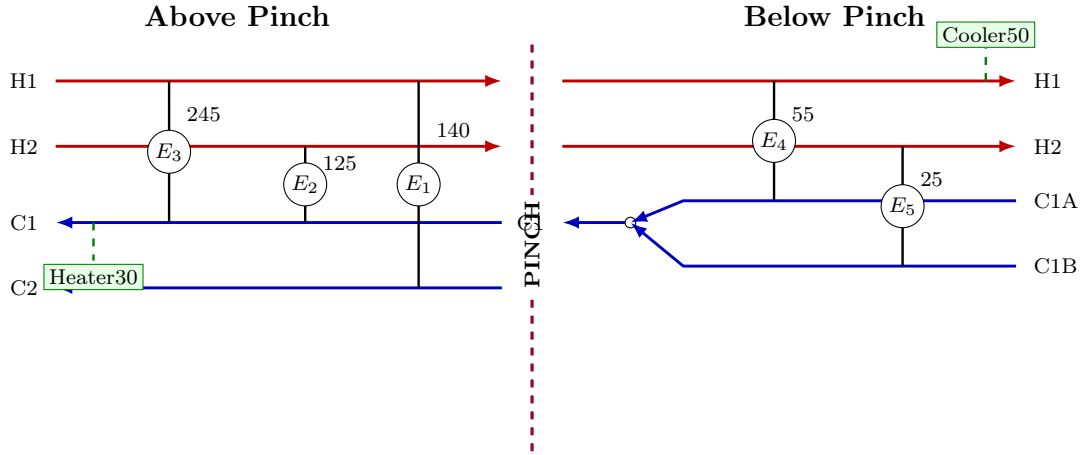


Figure 2: Schematic grid diagram for the pinch-compliant network. Numerical labels denote heat duties in kW.

The total internal heat recovery is

$$Q_{\text{recovered}} = 140 + 125 + 245 + 55 + 25 \quad (155)$$

$$= 590 \text{ kW}. \quad (156)$$

The same result is obtained independently from either side of the overall process balance:

$$Q_{\text{cold}} - Q_{H,\min} = 620 - 30 = 590 \text{ kW}, \quad (157)$$

$$Q_{\text{hot}} - Q_{C,\min} = 640 - 50 = 590 \text{ kW}. \quad (158)$$

Finally,

$$Q_{H,\min} + Q_{\text{hot}} = Q_{\text{cold}} + Q_{C,\min}, \quad (159)$$

and numerically,

$$30 + 640 = 620 + 50 = 670 \text{ kW}. \quad (160)$$

The proposed network therefore

- attains the minimum hot and cold utility targets,
- transfers no heat across the pinch,
- uses no cooling utility above the pinch,
- uses no heating utility below the pinch,
- satisfies the specified minimum temperature difference in every internal exchanger,
- and satisfies all process-stream target temperatures.

The example also demonstrates that the thermodynamic targets alone do not determine the network topology. Stream splitting may be required to construct a feasible network that realizes those targets.

4 The Workbook HEN_Pinch_Analysis_Helper.xlsm

The accompanying workbook `HEN_Pinch_Analysis_Helper.xlsm` was developed as an educational and validation tool. Its purpose is to support the learning process by automating the repetitive calculations used during a pinch study and by allowing small examples to be explored quickly.

The workbook assists the user throughout the thermodynamic targeting stage of the methodology. Starting from a small set of user inputs, it generates the required data structures, performs the pinch calculation, and creates a first grid representation of the process.

The workflow consists of four consecutive steps:

1. Specify the number of hot streams, the number of cold streams, and the minimum temperature difference $\Delta T_{\min}$.
2. Generate the stream-data tables and enter the process data for each stream. For every stream, the workbook calculates the heat-capacity flow rate

$$CP = \dot{m} c_p,$$

and automatically generates the shifted temperatures.

3. Perform the pinch analysis. The workbook constructs the shifted temperature levels, builds the temperature intervals, calculates the heat cascade, and determines the utility targets and pinch temperatures.
4. Generate a grid diagram that separates the process into the regions above and below the pinch and provides a starting point for network design.

The workbook is intended to support the stages described in Sections 1 and 2. It allows examples to be created, checked, and modified quickly, while keeping the individual calculation steps transparent.

The subsequent synthesis of a complete Heat Exchanger Network is not part of the current workbook. In particular, the workbook does not automatically perform stream matching, stream splitting, heat-exchanger selection, or network optimization. These steps remain engineering design tasks that must be carried out after the thermodynamic targets have been established.

Consequently, the workbook should be viewed as a compact learning and validation environment for pinch analysis rather than as a full heat-exchanger network synthesis package.